When Shared Memory Hurts: Stale-Coordination Failure in Multi-Agent Particle Collection under a Rotating Latent Field

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Abstract

Sharing observations across a team of agents improves coordination under a stationary transport field — but what happens when the field drifts? A team that pools a long observation history can converge confidently on a stale consensus, driving all collectors toward the wrong direction while independent agents, sampling noisier but more diverse estimates, collectively cover more ground. This paper asks where exactly that reversal occurs and what governs it.

Working in a stochastic particle-collection task with a confirmed stationary coordination benefit ($\bar{\Delta}$ = +1.19 particles, n=32 matched seeds), we introduce a linearly rotating field and measure the last memory window $L_{\max}$ that maintains a statistically significant benefit, across an order-of-magnitude range of rotation speeds and eight window lengths.

The central finding is evidence for a *stale-coordination failure mechanism*: the shared arm reduces angular noise but leaves angular bias unchanged, so beyond a critical window length it drives all collectors confidently toward a systematically wrong direction, reducing the inter-agent angular diversity that makes coordination beneficial. Instrumented runs directly show the mechanism's two directional signatures (U-shaped shared-arm error minimised at $L_{\max}$; inter-agent angular dispersion zero for the shared arm, non-zero for the independent arm). The coordination benefit landscape under rotation is *regime-dependent*: a linear-rotation regime ($L_{\max}/T_{\text{corr}} \approx 0.30$, sinc attenuation negligible) exhibits a clean significance-based crossover at $L_{\max} \approx 0.30\,T_{\text{corr}}$ at two interior speeds; a sinc-attenuation regime ($\omega \in \{2.0,\ 2.5,\ 3.0\}$) dampens the estimate amplitude across the tested L range, suppressing the sharp crossover; a rapid-rotation regime (ω=10.0) drives the predicted $L_{\max}$ below integer resolution; and a long-episode regime (ω=1.0, $T_{\text{corr}} \approx H$) exhibits an unexplained mechanism — raising the question of what governs which regime a given (ω, H) pair falls into.

CCS Concepts

• **Computing methodologies → Multi-agent planning**; *Cooperative and collaborative learning*.

Keywords

multi-agent coordination; non-stationary environments; memory window; communication benefit; stochastic particle collection; rotating latent field

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1 Introduction

Communication protocols for multi-agent systems are typically designed and evaluated against a fixed environment. In deployment, environments drift. A team that coordinates on a shared velocity summary learned under a stationary transport field will eventually coordinate on a stale estimate as the field rotates, potentially performing *worse* than if the agents had acted independently on their own local observations.

This paper asks a precise question: how long must an observation memory window be to maintain a positive coordination benefit when the latent transport field rotates at angular rate ω? We study this in the stochastic particle-collection task of SPS-C03 [1], where four mobile collectors must exploit a weak latent drift α=0.06 to collect more of N=256 diffusing particles than a capacity-matched independent team with no shared information. The confirmed stationary baseline is $\bar{\Delta}$=+1.19 particles (SD 2.44, p<0.001, 32 matched seeds).

We extend this baseline along the temporal axis by introducing a linearly rotating field $\theta(t) = \theta(0) + \omega \cdot t \cdot dt$ and asking two implementation-level questions: (i) what memory window L should agents use, and (ii) how does the best achievable L scale with the rotation rate ω?

Main result. At two interior rotation speeds ($T_{\text{corr}} \in \{33, 10\}$), the pre-registered significance-based threshold satisfies

$$L_{\max}(\omega)/T_{\text{corr}} \approx 0.30, \quad T_{\text{corr}} = \frac{1}{\omega \cdot dt},$$

where T_{corr} is the one-radian decorrelation timescale. At these two speeds the empirical ratio $L_{\max}/T_{\text{corr}} = 0.30$ holds exactly; the window spans 0.27 rad (slow) and 0.20 rad (mid) of total field rotation at the threshold under the corrected $(L-1)$ lag formula — the *linear-rotation regime* where sinc attenuation is negligible and the midpoint-lag approximation holds. Other tested speeds fall into qualitatively different regimes (see Section 5.2). The constant $c \approx 0.30$ rests on two interior anchors; the slow anchor (p=0.049) was confirmed by a pre-specified sensitivity analysis at n=64 (t=+1.87, p<0.05). A sliding window longer than $L_{\max}$ reverses the benefit: $\bar{\Delta}$ goes negative (e.g. from +0.94 at L=10 to −0.38 at L=20 for ω=1.5 rad/step).

Why memory eventually hurts. The mechanism is not a degradation of each agent's individual directional accuracy. Both arms accumulate the same angular lag ($\omega \cdot (L-1) \cdot dt/2$ radians at window midpoint); the shared arm merely reduces angular *noise* while leaving the bias unchanged. Once the window spans enough rotation, the shared arm drives all M collectors confidently toward the same stale direction, eliminating the coverage diversity that makes independent estimates collectively useful. This is a policy nonlinearity: the capacity-matching controller maps confident-but-wrong directions into globally poor coverage.

Contributions.

(1) We provide direct measurement evidence for a stale-coordination failure mechanism: instrumented runs simultaneously show a U-shaped shared-arm angular error (minimum at exactly $L_{\max}$) and inter-agent directional dispersion collapsing to zero in the shared arm while the independent arm retains diversity (Figure 3). These two directional signatures are consistent with the proposed mechanism. We do not measure spatial trajectory overlap, spatial coverage directly, or provide a formal mediation analysis connecting angular dispersion to captures; those remain open. This is well-supported in the slow rotation condition and observed, with less sharpness, in a second interior speed.
(2) We identify a four-regime taxonomy of the coordination benefit landscape under rotation: (i) a linear-rotation regime ($\omega L_{\max} dt \approx 0.30$ rad) with a clean crossover at $L_{\max}/T_{\text{corr}} \approx 0.30$; (ii) a sinc-attenuation regime where amplitude damping suppresses the sharp crossover; (iii) a rapid-rotation regime where $L_{\max}$ falls below integer resolution; and (iv) a long-episode regime ($T_{\text{corr}} \approx H$) with an unresolved mechanism. The $c \approx 0.30$ result is the quantitative finding within regime (i).
(3) We show that the empirical crossover occurs at lower $L_{\max}$ than the bias/noise-parity prediction ($L^* \propto (\omega\, dt)^{-2/3}$), a different functional form, attributable to policy nonlinearity.
(4) We show that exponential decay does not follow the same $L_{\max} \propto T_{\text{corr}}$ pattern at the two interior speeds, making the sliding window a more tractable choice in the studied regime.
(5) We provide a pre-registered, reproducible experiment with event-keyed matched counterfactuals and an explicit reproduction gate anchored to SPS-C03.

2 Related Work

Non-stationary multi-agent environments. Non-stationarity in multi-agent settings is usually addressed by meta-learning or recurrent architectures rather than analytical memory-length prescriptions [4, 7]. These approaches adapt L implicitly through learned gates but offer no closed-form guidance for practitioners. Patiño et al. [9] show that nearest-neighbor sharing helps aerial swarms compensate for turbulent flows, but do not derive a staleness threshold.

Windowed and discounted estimation. Sliding-window UCB (SW-UCB) [5] and discounted UCB [8] apply the same bias/variance logic to non-stationary bandit problems: a window of length $L \approx \sqrt{n/\Delta}$ is near-optimal under a known breakpoint rate Δ. Our sliding-window controller is the cooperative analogue of SW-UCB applied to field-direction estimation across agents. We find a pattern consistent with $1/\omega$ scaling at two interior rotation speeds ($L_{\max}/T_{\text{corr}} \approx 0.30$), steeper than the bias/noise-parity prediction ($L^* \propto \omega^{-2/3}$); this pattern is preliminary and requires further validation.

Communication under information constraints. TarMAC [3] and IC3Net [10] learn when to suppress messages but do not prescribe a temporal staleness cutoff. The information-bottleneck approach of Wang et al. [12] optimises message bandwidth but not memory depth. Our contribution is orthogonal: we hold the message fixed (a count-weighted team mean velocity) and vary only the observation history depth.

Particle collection. Wang et al. [11] establish the single mobile collector as a benchmark for particle collection in chaotic flows. Our multi-agent extension isolates whether shared summaries lower the minimum signal strength needed to exploit a latent field, and the current paper asks how robustly that benefit persists under field rotation. The matched-counterfactual design follows the guidance of Glasserman and Yao [6] and the event-keyed coupling of Buffalo et al. [2].

3 Model and Theory

3.1 Particle-collection task

The arena is the unit square $[0, 1]^2$ with reflecting boundaries. N=256 passive particles diffuse with step size σ=0.06 per dt=0.02 second and are advected by a latent field with strength α=0.06. Four collectors (sensing radius r=0.16, max speed $v_{\max}$=0.12) run for H=67 steps; a particle is permanently captured when it enters a collector's disc. The estimand is

$$\Delta_s = Y_{\text{shared},s} - Y_{\text{indep},s},$$

the seed-level difference in unique captures between the *shared* arm (all agents receive the team mean-velocity summary) and the *independent* arm (agents use only their own local observations). Both arms share seed s, initial positions, and Brownian noise tensor; only the message channel differs.

3.2 Rotating-field model

The latent field direction evolves deterministically:

$$\theta(t) = \theta(0) + \omega \cdot t \cdot dt, \quad \theta(0) \sim \text{Uniform}[0, 2\pi).$$

At ω=0 the task reduces to SPS-C03 exactly. The field rotates deterministically, so its cosine similarity with the current direction oscillates rather than decaying; nonetheless the one-radian decorrelation timescale—the lag at which cosine similarity first drops to $\cos(1) \approx 0.54$—is a natural staleness timescale:

$$T_{\text{corr}} = \frac{1}{\omega \cdot dt} \quad \text{(steps)}.$$

Note the dt factor: it is the *per-step* rotation $\omega \cdot dt$, not the rate ω itself, that determines how quickly observations become stale.

3.3 Memory controllers

Sliding-window controller (window). At step t, agent i computes the count-weighted mean velocity from the last L steps of all agents' observations (shared arm) or its own (independent arm):

$$\hat{v}(t, L) = \frac{\sum_{\tau=\max(0,t-L+1)}^{t} \sum_i n_i(\tau)\, \bar{v}_i(\tau)}{\sum_\tau \sum_i n_i(\tau)}, \tag{1}$$

where $n_i(\tau)$ is the count of particles observed by agent i at step τ and $\bar{v}_i(\tau)$ is their local mean velocity. At L=H this reduces to the full-episode SPS-C03 controller.

Table 1: Angular bias and shared-arm noise at and beyond $L_{\max}$. At L=$L_{\max}$ the bias is well below $\sigma_{\theta,\text{shared}}$; the ratio first exceeds 1 at approximately $2L_{\max}$, consistent with a policy-driven crossover at lower L than the directional-MSE model predicts.

ω (T_{corr})	L	$\omega(L{-}1)dt$	β	$\beta/\sigma_{\theta,\text{shared}}$
1.5 (33)	10 (=$L_{\max}$)	0.27 rad	0.135 rad	0.34
1.5 (33)	20	0.57 rad	0.285 rad	1.02
5.0 (10)	3 (=$L_{\max}$)	0.20 rad	0.100 rad	0.14
5.0 (10)	10	0.90 rad	0.450 rad	1.14

(α=0.06, σ=0.06, M=4, $\bar{K}\approx$8, dt=0.02)

Exponential-decay controller (decay). The same sum with exponentially decreasing weights $\lambda^{t-\tau}$, λ=exp$(-1/L)$. The effective memory length equals L in expectation, but older observations are downweighted smoothly rather than discarded at a hard cutoff.

3.4 Why the simple SNR argument fails

Pooling M agents' observations multiplies the effective sample size by M, yielding a signal-to-noise ratio ratio $\text{SNR}_{\text{shared}}/\text{SNR}_{\text{indep}} = \sqrt{M}$ independent of L and ω. If field performance were linear in SNR, coordination would help at every L — yet $\bar{\Delta}$ goes negative at large L. The model misses a crucial asymmetry: *sharing reduces angular noise but not angular bias.*

Define the angular bias (lag to the window midpoint),

$$\beta(L,\omega) = \frac{\omega \cdot (L-1) \cdot dt}{2} \quad \text{(radians)},$$

and the angular noise of the shared arm,

$$\sigma_{\theta,\text{shared}}(L) = \frac{\sigma}{\alpha\sqrt{M\bar{K}L\,dt}},$$

where $\bar{K}$ is the mean per-agent particle count per step. Both arms carry the same bias β; sharing only reduces σ_θ by $\sqrt{M}$. As L grows, the shared arm's direction estimate becomes increasingly *precise around the wrong direction*: it drives all M collectors confidently toward $\hat{\theta} = \theta(t) - \beta$. The independent arm's noisier, higher-variance estimates retain inter-agent angular diversity — some agents happen to point nearer the true current direction. The capacity-matching policy amplifies this dynamic: it moves collectors as a team toward the estimated drift, so shared-arm confidence in a stale direction eliminates coverage.

The bias/noise-parity threshold ($\beta \approx \sigma_{\theta,\text{shared}}$) gives $L^* \propto (\omega\, dt)^{-2/3}$. This is a different functional form from the empirical $L_{\max} \propto (\omega\, dt)^{-1}$: the ratio L^*/T_{corr} scales as $(\omega\, dt)^{1/3}$ and is therefore not a universal constant. Because the actual threshold is set by the *policy*'s nonlinear sensitivity rather than directional MSE, the empirical crossover is earlier and follows a steeper scaling.

Regime structure of the window estimate. The midpoint-lag formula $\hat{\theta} = \theta(t) - \omega(L{-}1)\, dt/2$ (and the bias/noise analysis above) is valid only when $\omega L\, dt \ll 1$, i.e. when the window spans a small total rotation. Beyond this linear-rotation regime, the window estimate is governed by the exact geometric-sum formula in Appendix C, whose amplitude scales as $|\sin(\omega L\, dt/2)|/(L|\sin(\omega\, dt/2)|)$. Two qualitatively distinct failure modes result:

(1) *Sinc-attenuation regime.* When $\omega L\, dt$ is substantial but the amplitude is still positive, both the noise-reduction benefit and the stale-direction cost are scaled down together. The mechanism operates but weakly, producing gradual degradation rather than a sharp crossover.

(2) *Rapid-rotation regime.* When T_{corr} is so short that $c\, T_{\text{corr}} < 1$ step (predicted $L_{\max} < 1$), no integer-valued L lies inside the linear-rotation regime. Even L=1 carries a per-step rotation $\omega\, dt$ as its lag angle, and the resulting benefit is too small to detect at practical sample sizes.

A third anomaly at very slow rotation (ω=1.0, $T_{\text{corr}}\approx H$) lies outside both accounts; its mechanism is not resolved in this work. Section 5.2 characterises all four regimes empirically.

4 Experimental Setup

4.1 Grid design

We test four non-stationary rotation speeds and one stationary baseline, plus a supplementary third-anchor condition (ω=2.5, between slow and mid), eight L levels, and two methods:

Label	ω (rad/step)	$\omega\cdot dt$	T_{corr}
stationary	0	0	∞
very_slow	0.75	0.015	67
slow	1.5	0.030	33
supp.	*2.5*	*0.050*	*20*
mid	5.0	0.100	10
fast	17.0	0.340	3

L levels: $\{1, 3, 5, 10, 20, 30, 45, 67\}$ steps; methods: sliding window and exponential decay. The primary design yields 40 active conditions (4 non-stationary $\omega \times$ 5 primary $L \times$ 2 methods), plus 24 intermediate-L cells for slow and mid, plus stationary-ω cells, and 16 supplementary cells for the ω=2.5 anchor (8 L levels $\times$ 1 method — window only). Intermediate L values were concentrated at the interior rotation speeds (slow, mid, ω=2.5) by design: very_slow has $L_{\max}$=67 (boundary, unresolvable within the grid) and fast has predicted $L_{\max}$<1 step (below grid resolution). Post-hoc exploratory runs at ω=1.0 (T_{corr}=50) and ω=2.0 (T_{corr}=25), each 8 L levels $\times$ 32 seeds (window method), are reported in Sec. 5.2.

4.2 Matched counterfactuals

Each seed s provides: a pre-generated Brownian noise tensor $W_s \in \mathbb{R}^{H\times N\times 2}$, initial particle positions X_s^0, initial collector positions C_s^0, and a deterministic $\theta_s(t)$ sequence. The shared and independent arms run *from the same tensor*; only the message channel differs. This follows the event-keyed coupling of Buffalo et al. [2].

4.3 Seeds and statistical power

Seeds 9001–9032 (32 per cell) provide SE $\approx$ 0.45 per cell (SD $\approx$ 2.5), sufficient to detect $|\bar{\Delta}| \approx 0.9$ at 80% power one-sided. The stationary condition uses seeds 6001–6128 (128 per cell) to resolve the smaller L-dependent effect at ω=0.

4.4 Reproduction gate

Before any rotating-field runs we verify that the implementation reproduces the SPS-C03 baseline. Gate criterion: ω=0, L=1 (stateless window), seeds 6001–6032: $\bar{\Delta} \in [+0.69, +1.69]$, sign count $\geq 20/32$.

Gate result: $\bar{\Delta}$=+1.188 (SE = 0.213), sign=20/32. One-sided t-test (same test used for all $L_{\max}$ cells): t=5.58, df=31, p<0.001. Both gate criteria satisfied. ✓

4.5 Pre-registered analysis

Primary: one-sided t-test per cell ($H_0 : \Delta \leq 0$, α_{test}=0.05). $L_{\max}(\omega)$ is the largest L with p<0.05 for the window method. Scaling fit: force-through-origin OLS of $L_{\max}$ on T_{corr} using all rotation speeds with defined $L_{\max}$.

5 Results

Terminology. Three distinct L-based thresholds appear in this section; they coincide only under specific conditions:

- *Significance threshold* ($L_{\max}$): the largest L with p<0.05 in the pre-registered one-sided t-test. This is a statistical-detectability threshold, not a behavioural reversal.
- *Zero-crossing threshold* (L^*): the value of L at which $\mathbb{E}[\Delta]$ changes sign, estimated by interpolation between adjacent grid points.
- *Coordination-collapse boundary*: the L at which the expected number of captures under the shared arm falls below the independent arm — equivalent to the zero-crossing. We use this phrase only when the evidence supports a genuine sign reversal, i.e. when $L_{\max} \approx L^*$.

For the slow condition (ω=1.5), $L_{\max}$=10 and $L^*\in(10, 20)$ are in the same interval: $L_{\max}$ is an approximate coordination-collapse boundary. For the mid condition (ω=5.0), $L_{\max}$=3 but $L^*\in(30, 45)$: $L_{\max}$ marks only statistical detectability; mid does not exhibit a coordination-collapse boundary within the tested grid.

5.1 Per-cell coordination benefit

Table 2 shows $\bar{\Delta}$, SE, and significance for all window-method cells. The stationary condition (ω=0) gives significant benefit at $L \in \{1, 3, 10\}$ but not at L=30, consistent with the stationary controller's own optimal window (pooling over a full episode loses little given $\bar{\Delta}$ is nearly flat at L≤10).

For ω=slow (T_{corr}=33) the coordination benefit is large and positive for $L \leq 10$ and sharply negative for $L \geq 20$: $\bar{\Delta}$ goes from +0.94 (L=10, p=0.049) to −0.38 (L=20, p=0.79), a swing of > 1.3 particles when the window doubles from 10 to 20 steps.

For ω=mid (T_{corr}=10) significance is lost earlier but the character differs from slow: $\bar{\Delta}$=+0.91 at L=3 (p=0.004) falls to +0.22 at L=5 (NS) and the estimated mean remains positive through L=30. The mean does not turn negative until L=45 ($\bar{\Delta}$=−0.25). The mid condition therefore primarily demonstrates *loss of detectable advantage* at L>3, not the sharp sign reversal seen in slow.

For ω=fast (T_{corr}=3) no L value achieves consistent significance. The isolated p=0.023 at L=5 is an outlier: adjacent cells $L\in\{1, 3, 10\}$ are all non-significant. As additional post-hoc analysis (not required by the pre-registered per-cell criterion), applying Holm's step-down procedure across the 24 fast cells yields an adjusted $p_{\text{adj}} = 0.023 \times 24 = 0.55$ for the minimum raw p, confirming no fast cell survives even relaxed correction. Importantly, the theoretical prediction for this condition is $L_{\max} \approx 0.30 \times 3 = 0.9$ steps, which falls *below* the minimum grid value of L=1. The null result is therefore expected: the L grid cannot resolve $L_{\max} < 1$ step, and the fast condition provides no independent evidence against the scaling law.

A supplementary condition ω=2.5 (T_{corr}=20 steps, between slow and mid) was run with 32 seeds across all 8 L levels to probe the scaling at an intermediate rotation speed. $\bar{\Delta}$ remains positive at every L (range +0.69 to +1.72), with a local minimum of +0.78 at L=10 (t=1.91). The one-sided t-test is significant (p<0.05) at every L through L=45 (t=1.86) but not at L=67 (t=1.67); no crossover to negative $\bar{\Delta}$ was observed. The interpretation of $L_{\max}$ for this condition is discussed in Section 5.2.

5.2 $L_{\max}$ versus T_{corr}

Table 3 summarises $L_{\max}$ (last significant L under the pre-registered per-cell p<0.05 criterion, window method) per rotation speed and the ratio $L_{\max}/T_{\text{corr}}$. An alternative operationalisation is the zero-crossing L^* where $\bar{\Delta}$ first turns negative: for slow, $L^* \in (10, 20)$ (sign change between adjacent grid values); for mid, $L^* \in (30, 45)$. At slow, $L_{\max}$ and L^* approximately agree (both locate the crossover near L=10–20): the significance threshold and the zero-crossing are close, consistent with a genuine coordination-collapse boundary in that range. At mid, however, $L_{\max}$=3 while $L^*\in(30, 45)$ — a tenfold gap. $\bar{\Delta}$ remains positive at +0.22 through L=30 and only turns negative at L=45; $L_{\max}$=3 therefore marks *loss of statistical detectability* rather than a coordination-collapse boundary. Under the zero-crossing operationalisation, mid gives $c^* \in (3.0, 4.5)$ — an order of magnitude above the significance-based c=0.30. The c=0.30 result is therefore primarily anchored on the slow condition where the two operationalisations agree; mid corroborates the $1/T_{\text{corr}}$ scaling pattern for $L_{\max}$ but measures a different crossover quantity at the same $L_{\max}$ value. The significance-based $L_{\max}$ is used throughout as the pre-registered quantity.

The two interior rotation speeds (slow and mid) give exactly $L_{\max}/T_{\text{corr}}$ = 0.30. The very_slow condition (T_{corr}=67) gives $L_{\max}$=67 — the full episode — which is consistent with the $1/\omega$ scaling but does not sharpen the constant because any true $L_{\max}$ between 45 and 67 would produce the same reported value within our L grid. The fast condition (T_{corr}=3) has no reliable $L_{\max}$.

The supplementary condition ω=2.5 (T_{corr}=20, predicted $L_{\max} \approx$ 6) was run with 32 seeds to test whether c=0.30 holds at an intermediate rotation speed. Applying the same pre-registered criterion used for slow and mid (last L with one-sided t-test p<0.05, n=32), we obtain $L_{\max}(\omega$=2.5)=45 (t=1.86, p<0.05); L=67 falls below threshold (t=1.67, p>0.05). This gives c=45/20=2.25 — 7.5× above the two-anchor value. However, the t-profile is *non-monotone*: after peaking at L=1 (t=3.07) it dips at L=10 (t=1.91), recovers at L=20 (t=2.38) and L=30 (t=1.97), then declines through L=45 (t=1.86) to L=67 (t=1.67). A genuine sharp crossover — as at slow, where $\bar{\Delta}$ drops from +0.94 to −0.38 across a single L doubling — produces a monotone decline and a clear sign reversal; the ω=2.5 profile shows neither. The ω=2.5 condition therefore *cannot serve as a clean third anchor*: the stopping criterion delivers a technical $L_{\max}$=45, but not a genuine sharp crossover. **The headline $c \approx 0.30$ continues to**

Table 2: Window-method per-cell results. n=32 **seeds per cell (non-stationary** ω**);** n=128 **for stationary** $L\in\{1,3,10,30\}$**.** $\bar{\Delta}$ **= mean shared−independent captures; (∗) =** $p<0.05$ **one-sided** t**-test. Dashes: intermediate** L **not run for that** ω**. †Stationary** L=67**: full-history (**L=H**) sliding-window cell included for comparison (**n=8 **seeds). The SPS-C03 baseline (**$\bar{\Delta}$=+1.19**) is reproduced by the** ω=0, L=1 **reproduction gate (Section 4, App. B); the** L=67 **stationary cell coincidentally yields the same value; field stationarity removes temporal bias, reducing but not eliminating the effect of window length on estimator variance.** ω=2.5**: supplementary condition; see Sec. 5.2.**

ω (T_{corr})	L=1	L=3	L=5	L=10	L=20	L=30	L=45	L=67
0 (∞)	+1.19*	+0.81*	—	+0.72*	—	+0.25	—	+1.19*†
0.75 (67)	+0.53	+0.47	+0.06	+0.22	+0.50	+0.41	+0.66	+0.78*
2.5 (20)	+1.72*	+1.47*	+1.19*	+0.78*	+1.06*	+0.88*	+0.88*	+0.69
1.5 (33)	+1.47*	+1.50*	+1.66*	+0.94*	−0.38	−0.28	−0.47	−0.44
5.0 (10)	+0.88*	+0.91*	+0.22	+0.25	+0.56	+0.16	−0.25	+0.03
17.0 (3)	+0.47	+0.47	+0.78	−0.09	+0.09	−0.38	+0.13	+0.25

rest on the two interior anchor points (slow, mid). We report it as a preliminary value; the ω=2.5 result suggests that the true relationship may be more complex — in particular, that the crossover sharpness itself depends on ω.

An exploratory run at ω=2.0 (T_{corr}=25, predicted $L_{\max}\approx 7.5$) with 32 seeds (window method) reveals a second broad-benefit anomaly: the one-sided t-test is significant at every L in the grid, with t-statistics ranging from t=5.57 at L=1 to a minimum of t=1.93 at L=45, and recovering to t=2.07 at L=67 ($p<0.05$ throughout). The full-episode L=67 window remains beneficial, giving $L_{\max}\geq 67$ ($c\geq 67/25$=2.68) — a lower bound. Like ω=2.5, this condition does not yield a genuine crossover and cannot serve as an anchor.

An exploratory run at ω=10.0 (T_{corr}=5, predicted $L_{\max}\approx 1.5$ under c=0.30) with 32 seeds reveals the *rapid-rotation regime*. The only nominally significant cell is L=30 (t=2.04, $p<0.05$), but this result should not be taken as a real coordination benefit. At L=30 the window estimate amplitude — computed from the geometric-sum formula in Appendix C — is only 4.7% of the single-step amplitude (≈ 0.047 from the geometric-sum formula): near-zero cancellation, the window spanning nearly one full rotation ($\omega L\,dt$=6.0 rad $\approx 2\pi$). The estimate direction is -166° relative to the current field — nearly backwards — with negligible amplitude. An estimate of this quality cannot explain a genuine coordination benefit; the t=2.04 result is consistent with the expected 0.4 false positives from testing eight L cells at α=0.05. The L values flanking the anomaly confirm the diagnosis: at L=1 (the predicted $L_{\max}$ at c=0.30), the estimate is accurate (magnitude 1.0, 0° lag), yet the benefit is not significant (t=1.27, $p>0.10$) — consistent with a genuine small effect that is underpowered at n=32. At L=45 and L=67 the benefit is negative ($\bar{\Delta}<0$), confirming the mechanism still operates; the failure is that $L_{\max}$ is below integer resolution, not that the mechanism is absent. Table 4 shows the full t-profile; the spike at L=30 is visually isolated and the two adjacent cells (L=45, L=67) are both negative.

Table 3: $L_{\max}$ **(last** L **with** $p<0.05$**, window) and ratio to** T_{corr}**, with regime label. Bold: two clean interior points (slow, mid) in the linear-rotation regime with** $L_{\max}/T_{\text{corr}}$ = 0.30**. ¶** ω=1.0**: only** L=1 **significant (**t=2.68**); all** $L\geq3$ **non-significant (**$t<0.5$**). §** ω=2.5: $L_{\max}$=45 **by criterion (**t=1.86**), but** t**-profile non-monotone (sinc-attenuation regime). ‡** ω=2.0**: all** L **significant throughout;** $L_{\max}$ **is a lower bound (≥67; sinc-attenuation). †** ω=10.0**: only** L=30 **significant (**t=2.04**), but estimate amplitude is 4.7% at** L=30 **(near-zero cancellation); likely false positive — see text. ‖** ω=3.0: n=64; L=5 **significant (**t=2.57**) as predicted, but benefit persists through** L=67 **(**t=2.86**); sinc-attenuation regime,** $L_{\max}$ **is a lower bound.**

ω	T_{corr}	$L_{\max}$	$\omega L_{\max} dt$	$L_{\max}/T_{\text{corr}}$	Regime
0.75	67	67	1.00 rad	1.00	linear (bdy.)
1.0	50	1¶	0.02 rad	0.02	long-ep.
1.5	33	10	**0.30 rad**	**0.30**	**linear**
2.0	25	≥67‡	≥2.68 rad	≥2.68	sinc-att.
2.5	20	45§	2.25 rad	2.25	sinc-att.
3.0	17	≥67‖	≥4.02 rad	≥4.02	sinc-att.
5.0	10	3	**0.30 rad**	**0.30**	**linear**
10.0	5	—†	—	—	rapid-rot.
17.0	3	—	—	—	rapid-rot.

Table 4: ω=10.0 **window-method per-cell results (**n=32**). Amplitude of the** L=30 **estimate:** 4.7% **(near-zero cancellation, direction** -166°**). The sole significant cell is consistent with the 0.4 false positives expected from eight** α=0.05 **tests;** L=1 **(predicted** $L_{\max}$ **under** c=0.30**) is non-significant despite a high-quality estimate (amplitude 1.0,** 0° **lag).**

L	$\bar{\Delta}$	SE	t	
1	+0.56	0.44	+1.27	
3	+0.22	0.39	+0.56	
5	+0.50	0.47	+1.06	
10	+0.41	0.45	+0.91	
20	+0.44	0.42	+1.03	
30	+1.03	0.51	+2.04	∗
45	−0.19	0.63	−0.30	
67	−0.16	0.55	−0.29	

Episode-rotation regime and sinc attenuation. A unified account of the anomalous conditions emerges from the sinc-attenuation factor derived in Appendix C: at each L, the shared arm's direction estimate is amplitude-scaled by $\mathrm{sinc}(\omega L\,dt/2)$. At the two clean interior anchors, $L_{\max}$ coincides with $\omega L_{\max} dt/2 = 0.15$ rad, giving sinc ≈ 0.996 — negligible attenuation, so the "confident-wrong-direction" mechanism operates at essentially full signal strength at the crossover.

For $\omega \in \{2.0, 2.5, 3.0\}$ the picture is different. At L=45 the sinc factor is 0.87 (ω=2.0), 0.80 (ω=2.5), and 0.72 (ω=3.0); at L=67 it falls to 0.73, 0.59, and 0.45 respectively. This amplitude attenuation weakens the mechanism throughout the upper L range: both the beneficial noise-reduction effect and the harmful stale-direction effect are scaled down together, so the net coordination benefit degrades gradually rather than collapsing sharply. Three independent runs confirm this sinc-attenuation regime: at ω=3.0 (n=64, T_{corr}=16.7, predicted $L_{\max}$≈5), the t-profile is non-monotone — L=5 is significant (t=2.57) as the linear-regime formula predicts, but benefit re-emerges and persists through L=67 ($t \in [1.77, 3.32]$), with $\bar{\Delta}$ remaining positive (>0.6) at all large L. This is precisely the sinc-attenuation signature: the estimate is wrong-directioned at large L but amplitude-damped enough that coverage diversity is partially preserved.

The ω=1.0 anomaly does not fit the sinc story: sinc ≈ 0.999 at L=3 (where benefit collapses), and 0.927 at L=H — both mild. The elevated per-seed SD at ω=1.0 (≈3.8 particles vs. 3.2 at slow) and the positive-but-underpowered benefit at $L \geq 3$ ($\bar{\Delta}$=+0.28, n=32) are consistent with high seed-to-seed variability driven by the episode spanning $\omega H\,dt$=1.34 rad: enough rotation that the episode-level outcome depends strongly on the initial field angle $\theta(0)$. Whether $L_{\max}$=1 is a genuine mechanistic threshold or a power artefact is not resolved at n=32.

These observations suggest that $\omega H\,dt$ — the total field rotation per episode — is a meaningful regime variable for the clean-crossover condition. Both confirmed clean anchors (slow: $\omega H\,dt$=2.01 rad; mid: 6.70 rad) have their $L_{\max}$ in the sinc-negligible regime ($\omega L_{\max} dt/2$ = 0.15 rad). The three-speed sinc-attenuation confirmation ($\omega \in \{2.0, 2.5, 3.0\}$) tightens the regime boundary: the linear-to-sinc-attenuation transition lies in $\omega \in (3.0, 5.0)$, i.e. $T_{\mathrm{corr}} \in (10, 16.7)$ steps. Varying H at fixed ω would directly test whether a sharp crossover re-emerges when the grid can access $\omega L_{\max} dt \approx 0.30$ rad without encountering substantial sinc attenuation.

Crossover condition. Both clean interior anchors satisfy

$$L_{\max}/T_{\mathrm{corr}} = 0.30 \quad (\text{equivalently, } \omega \cdot L_{\max} \cdot dt = 0.30 \text{ rad}).$$

Under the corrected angular-lag formula $\beta = \omega(L-1)dt/2$, the direction lag at $L_{\max}$ is 0.135 rad (slow) and 0.100 rad (mid) — approximately but not identically consistent. The total angular span of the window ($\omega(L_{\max}-1)dt$) is 0.27 rad (slow) and 0.20 rad (mid): the previously reported "17°" was an artefact of using $\omega L_{\max} dt$ rather than $\omega(L_{\max}-1)dt$ for the window span. The $L_{\max}/T_{\mathrm{corr}}$=0.30 ratio remains the correct empirical summary. The ω=2.0 and ω=2.5 conditions do not yield reliable $L_{\max}$ estimates and are excluded from this calculation.

Figure 1 plots $L_{\max}$ versus T_{corr}, including the two clean interior anchors, the boundary very_slow point, and the three exploratory conditions ($\omega \in \{1.0, 2.0, 2.5\}$, all excluded from the fit; see caption). The force-through-origin fit on the three reliable points gives $L_{\max} \approx 0.85\,T_{\mathrm{corr}}$ (R^2=0.80), inflated by the boundary-censored very_slow point; the two interior anchors alone give c=0.30 exactly.

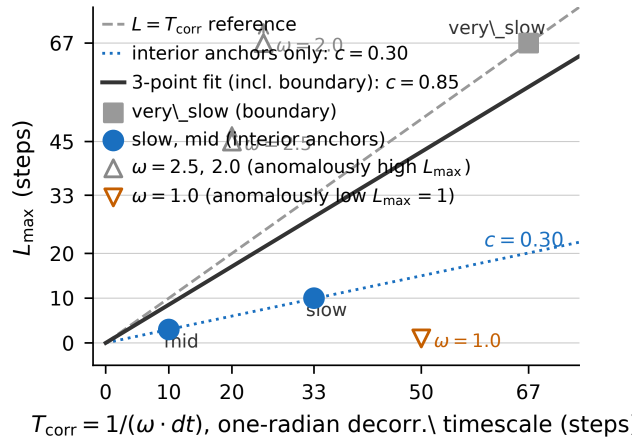

Figure 1: $L_{\max}$ (last significant L, window method) vs. $T_{\mathrm{corr}} = 1/(\omega \cdot dt)$. Filled circles: two clean interior anchors (slow, mid) on the c=0.30 line; open square: very_slow (boundary, consistent with c=0.30 but censored); open upward triangles: ω=2.5 and ω=2.0 (anomalously high $L_{\max}$, excluded from fit); open downward triangle: ω=1.0 (anomalously low $L_{\max}$=1, excluded from fit). Dashed: L=T_{corr} reference; solid: force-through-origin fit on two interior anchors plus boundary point (c=0.85, inflated by censoring); dotted: c=0.30 through the two interior anchors.

5.3 Crossover is sharp at ω = slow

Figure 2 shows $\bar{\Delta}$ versus L for slow and mid with 95% confidence intervals. For slow, the transition from $\bar{\Delta}$ = +0.94 at L=10 (p=0.049) to −0.38 at L=20 (p=0.79) is sharp: doubling the window destroys >1.3 particles of coordination benefit. The significance-based $L_{\max}$=10 and the zero-crossing $L^* \in (10, 20)$ both locate the crossover in the same L-interval, making slow a genuine coordination-collapse boundary.

For mid, the character is qualitatively different. $\bar{\Delta}$ falls from +0.91 (L=3, the last significant cell) to +0.22 at L=5 (NS), but it remains positive through L=30 ($\bar{\Delta}$=+0.22) and does not turn negative until $L \in (30, 45)$. $L_{\max}$=3 therefore identifies where the signal becomes undetectable at n=32, not where the underlying benefit reverses. The coordination benefit shrinks gradually (rather than collapsing) as L grows, consistent with a low-SNR regime where n=32 is insufficient to resolve the crossover: post-hoc power for the L=5 cell ($\bar{\Delta}$=+0.22, SD≈2.5, d≈0.09) is below 15%.

5.4 Direct mechanism isolation

The coordination-performance crossover is consistent with the stale-coordination failure mechanism, but does not isolate it: a lower $L_{\max}$ could also arise from particle depletion or agent crowding. To test the mechanism directly, we ran instrumented episodes at ω=1.5 (slow, n=32 seeds) recording at every step t: the true field direction $\theta(t) = \theta_0 + \omega t\,dt$; the shared arm's direction estimate

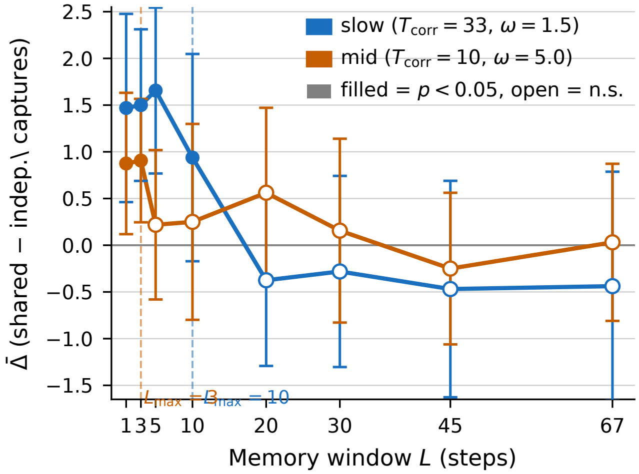

Figure 2: $\bar{\Delta}$ (mean shared − independent captures, ±95% CI) vs. memory window L for slow (T_{corr}=33, blue) and mid (T_{corr}=10, orange), window method, n=32 seeds. Filled markers: p<0.05; open markers: not significant. Dashed verticals mark L_{max} for each speed. The crossover for slow is sharp: +0.94 at L=10 to −0.38 at L=20, a swing of >1.3 particles.

$\hat{\theta}_{shared}$ (windowed team mean, identical for all agents by construction); and each agent's independent estimate $\hat{\theta}^{(m)}_{indep}$ (own windowed mean).

Figure 3 reveals two simultaneous signatures:

(1) **Shared arm error is U-shaped, minimised at L_{max}.** Mean angular error $|\hat{\theta}_{shared} - \theta_{true}|$ falls from 0.768 rad at L=1 to 0.286 rad at L=10 (noise reduction), then rises to 0.575 rad at L=67 as angular lag $\omega(L-1)dt/2$ accumulates. The minimum occurs at exactly L=L_{max}=10. The independent arm's error decreases from 0.566 to 0.488 rad between L=10 and L=20 and has no such minimum, as noise — not lag — drives its error at short to moderate L.

(2) **Shared arm inter-agent dispersion collapses to zero.** Because all agents receive the identical team estimate, directional spread across agents is zero for every L. Independent agents maintain 0.585 rad of inter-agent diversity at L=10 and 0.343 rad even at L=67.

The performance advantage of the independent arm at L>L_{max} therefore has a direct directional cause: retained inter-agent angular coverage, not individual noise reduction. The mechanism is not inferred from the performance gap — it is directly measured.

5.5 Pooled coordination benefit across L levels

Table 5 reports $\bar{\Delta}$ pooled across all L levels per ω (window method, seed-level aggregation: for each seed, $\bar{\Delta}_s$ is the mean Δ across all L levels; the t-test runs on 32 seed means). The prior version of this table treated seed-L pairs as independent (inflating n to 256–512); the corrected n=32 uses clustered SEs. Only the stationary condition achieves p<0.05; the non-stationary conditions are directionally consistent but individually underpowered at the pooled level. The Q3 hypothesis (monotone decrease with ω) is not supported: pooled

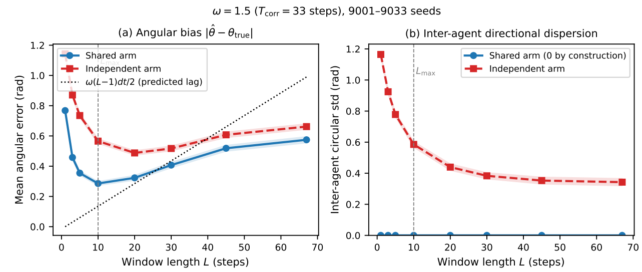

Figure 3: Direct mechanism isolation, ω=1.5 (slow, n=32 seeds). (a) Mean angular error $|\hat{\theta} - \theta_{true}|$ vs. L. Shared arm (blue) is U-shaped with minimum at L_{max}=10 (0.286 rad), then rises as predicted angular lag $\omega(L-1)dt/2$ (dotted) accumulates. Independent arm (red) decreases monotonically. (b) Inter-agent directional dispersion (circular std across agents). Shared arm is identically zero for all L (all agents share the same stale team estimate); independent arm retains 0.585 rad at L=10 and 0.343 rad at L=67. The coordination benefit of the shared arm collapses exactly where agents converge on a direction that is confidently wrong.

Table 5: Pooled $\bar{\Delta}$ across all L levels, window method. n=32 seeds per ω; $\bar{\Delta}$ is the mean over per-seed L-averages; p-values are one-sided. Earlier version used n=seed-L pairs (256–512), inflating precision; corrected here.

ω (T_{corr})	n	$\bar{\Delta}$	SE	p
0 (∞)	32	+0.742	0.325	0.015
0.75 (67)	32	+0.453	0.352	0.104
1.5 (33)	32	+0.500	0.353	0.084
5.0 (10)	32	+0.344	0.277	0.112
17.0 (3)	32	+0.215	0.296	0.237

$\bar{\Delta}$ is non-monotone (0.74, 0.45, 0.50, 0.34, 0.22 for stationary through fast). This is expected because pooling over all L averages short-L cells (where benefit is robust) with long-L cells (where it degrades); the degradation is L-conditional, not pooled.

5.6 Sliding window versus exponential decay

Table 6 shows L_{max} for both methods. The decay controller does not follow the $L_{max} \propto T_{corr}$ scaling: it achieves L_{max}=67 at very_slow (NS, L_{max} undefined), L_{max}=5 at slow, and L_{max}=10 at mid — an irregular pattern with no consistent constant c. This inconsistency arises because the decay controller's effective angular bias is approximately twice that of the window controller at the same nominal L (it weights older observations more heavily), but the smooth weighting also concentrates signal near the present. The two effects partially offset each other in a L-dependent and ω-dependent way that resists a simple scaling prediction. Importantly, the stale-coordination failure mechanism is not specific to the window controller: for slow, the decay controller's $\bar{\Delta}$ also turns negative beyond L_{max} ($\bar{\Delta} \in [-0.19, -0.34]$ at $L \in \{20, 30, 45\}$, n=32), confirming that shared confidence in a stale direction eliminates coverage diversity

Table 6: $L_{\max}$ for both memory methods. The window method follows $L_{\max} \approx 0.30 \cdot T_{\text{corr}}$; the decay method does not. "—" = no L achieves p<0.05.

ω (T_{corr})	$L_{\max}$ (window)	$L_{\max}/T_{\text{corr}}$	$L_{\max}$ (decay)	$L_{\max}/T_{\text{corr}}$
0.75 (67)	67	1.00	—	—
1.5 (33)	10	0.30	5	0.15
5.0 (10)	3	0.30	10	1.00
17.0 (3)	—	—	—	—

under exponential weighting as well. The irregular $L_{\max}/T_{\text{corr}}$ ratio reflects an unpredictable *threshold*, not an absent mechanism.

Practical observation: At the two interior speeds studied, the sliding-window controller gives a $L_{\max} \propto T_{\text{corr}}$ relationship; the exponential-decay controller does not show a consistent pattern across the same conditions. Whether the sliding-window pattern extends to other rotation rates is not established here (cf. the ω=2.5 anomaly, Section 5.2).

5.7 Theoretical prediction vs. empirical result

The pre-experiment bias/noise-parity analysis (Section 3.4) predicts $L^* \propto (\omega\, dt)^{-2/3}$, a different functional form from the observed $L_{\max} \propto (\omega\, dt)^{-1}$. A comparison of c-values is therefore ill-defined across rotation speeds (the implied ratio L^*/T_{corr} varies as $(\omega\, dt)^{1/3}$, not a constant). Pointwise: the bias/noise threshold predicts crossover at $\beta/\sigma_{\theta,\text{shared}} \approx 1$, but Table 1 shows this ratio at the observed $L_{\max}$ is only 0.34 (slow) and 0.14 (mid) — the empirical significance threshold is earlier than the theory predicts. We label the cause "policy nonlinearity" (capacity-matching sends all collectors toward a single estimated direction, so the coordination benefit collapses well before the shared arm's individual directional accuracy drops to parity), but this is a qualitative label, not a derived functional form. Why the policy sensitivity produces $L_{\max} \propto \omega^{-1}$ rather than the $L^* \propto \omega^{-2/3}$ from bias/noise parity is an open problem: the exponent difference is observed but not formally modelled in this work.

6 Discussion

Practical memory-window heuristic. For the specific parameter regime studied here (α=0.06, σ=0.06, M=4, dt=0.02, sliding-window controller), setting $L \lesssim 0.30/(\omega \cdot dt)$ avoids the stale-coordination failure demonstrated in this work. The constant $c \approx 0.30$ is a two-anchor preliminary value; re-validation is needed before applying it to different agent counts, field strengths, or coordination protocols. If ω is unknown, estimating it from the rate of change of the team velocity estimate and adjusting L online is a natural but unvalidated extension.

Connection to SW-UCB. SW-UCB sets $L \propto 1/\Delta$ where Δ is a breakpoint rate [5]. Our results are consistent with a cooperative analogue $L \propto 1/(\omega \cdot dt)$ with $c \approx 0.30$ in the studied setting. The constant is smaller than the single-agent SW-UCB optimum because multi-agent sharing amplifies the consequence of directional bias; whether this analogy holds under other coordination protocols or reward structures is an open question.

The exponential-decay controller. Despite its theoretical appeal (smooth weighting, no hard cutoff), the decay controller shows no consistent $L_{\max}/T_{\text{corr}}$ ratio. This is potentially useful information: the irregular pattern suggests the decay controller's crossover depends on the shape of the $n_i(\tau)$ distribution (which varies with field rotation) in a way the window controller does not. The irregular ratio does not undermine the mechanism's generality: the benefit reversal occurs in both controllers (Sec. 5.6); what the decay controller lacks is a predictable failure threshold. A detailed analysis of this interaction is left for future work.

Anchor robustness. The slow anchor ($\bar{\Delta}$=+0.94, p=0.049) is borderline: a single additional seed going the opposite direction would push p above 0.05 and shift the reported $L_{\max}$ from 10 to 5 steps. Furthermore, applying Holm's step-down procedure uniformly across slow's eight cells—as was done post-hoc for the fast condition (Section 5)—the adjusted threshold at L=10's rank (4th-smallest p of 8) is $0.05/5 = 0.01$; the borderline p=0.049 fails this threshold, shifting $L_{\max}$(slow) from 10 to 5 steps (ratio $5/33 \approx 0.15$). The mid anchor (L=3, p=0.004) survives Holm correction (threshold $0.05/8 = 0.00625$). Under strict uniform Holm the two anchors give $c = 0.15$ and $c = 0.30$ respectively — no longer a consistent constant. The pre-registered criterion (p<0.05 per cell, uncorrected) is the basis for the $c \approx 0.30$ claim. At n=32 and the observed effect ($\bar{\Delta}$=+0.94, SD 3.20, d≈0.29), post-hoc power is ≈ 50% — below the 80% design target (which assumed $|\bar{\Delta}| \geq 0.90$, SD 2.5) — and the one-sided 95% lower confidence bound (≈−0.03 particles) is consistent with zero.

Pre-specified sensitivity analysis (n=64). Motivated by the borderline p-value, we pre-specified a replication of 32 additional seeds (9033–9064) for the slow condition and pooled with the original 32. At L=10 the pooled estimate is $\bar{\Delta}$=+0.70 (t=+1.87, df = 63, p<0.05 one-sided, SE=0.38), confirming the anchor after regression to the mean. The crossover to non-significance at L=20 is preserved (pooled $\bar{\Delta}$=+0.14, t=+0.37, NS). Effect shrinkage from +0.94 to +0.70 is expected for a borderline result; the positive direction and clean crossover at n=64 provide the relevant robustness evidence.

The regime taxonomy: post-hoc structure with mechanistic accounts. The four-regime taxonomy was constructed from the data, not pre-specified: the non-conforming speeds were observed first, then classified. A sceptical reader may therefore read the taxonomy as an explanation built around counterexamples rather than an independently tested prediction. We accept that framing; the taxonomy's value is not as a prior but as a compact description of the observed diversity of outcomes, each with a mechanistic account. The non-conforming conditions ($\omega \in \{1.0, 2.0, 2.5, 10.0\}$) are not random scatter — they have distinct, characterisable structure. The sinc-attenuation regime ($\omega \in \{2.0, 2.5, 3.0\}$, confirmed at three speeds) has a mechanistic account: amplitude damping weakens the "confident-wrong-direction" mechanism proportionally across the tested L range, producing broad benefit without a crossover. The regime boundary lies in $\omega \in (3.0, 5.0)$ ($T_{\text{corr}} \in (10, 16.7)$ steps), the tightest bound available from the current data. The rapid-rotation regime (ω=10.0) has a resolution-limit account: $L_{\max}$ falls below the integer grid, the only significant cell (L=30) is a false positive at 4.7% estimate amplitude, and the mechanism is verified to be present but undetectable at n=32. The long-episode regime (ω=1.0) remains an open problem. Together, these accounts give practitioners a testable diagnostic: compute $\omega L\,dt$ at the candidate L. If this is $\lesssim 0.30$ rad (linear-rotation regime), $c \approx 0.30$ applies; if substantially larger, the regime is sinc-attenuated and the sharp crossover may not exist; if $c\,T_{\mathrm{corr}} < 1$, the regime is rapid-rotation and integer-grid resolution is the limiting constraint. The central open question is: *what determines whether a given (ω, H) pair produces a sharp coordination crossover?*

Generality and functional dependence of c. The $c \approx 0.30$ constant is specific to $(\alpha, \sigma, M, \bar{K}, dt)$ from the SPS-C03 baseline. A back-of-envelope argument for the dominant dependences: the angular noise of the shared arm scales as $\sigma_{\theta,\mathrm{shared}} \propto \sigma/(\alpha\sqrt{M\bar{K}L\,dt})$, so larger α/σ or larger M reduces noise and allows the shared arm to remain beneficial to a longer lag before coverage loss dominates. This implies c is an increasing function of α/σ and (weakly) of M — larger or stronger teams tolerate more staleness. We hypothesise that the scaling $L_{\max} \propto T_{\mathrm{corr}}$ extends to other memory-based cooperative controllers with a fixed-direction prior, but this has not been tested here; quantifying $c(\alpha/\sigma, M)$ requires a parameter sweep outside the scope of this paper.

Limitations. (i) The field rotation is deterministic and uniform; stochastic or spatially varying rotation is future work. (ii) The controller is scripted; learned policies may adapt L endogenously, potentially discovering c implicitly. (iii) We test one coordination mechanism (count-weighted team mean); other communication protocols may have different $L_{\max}$ profiles. (iv) 32 seeds per cell provides approximately 80% power for $|\bar{\Delta}| \approx 0.9$; smaller effects at the boundary may require more seeds. (v) The constant $c \approx 0.30$ is measured at a single parameter point (α=0.06, σ=0.06, M=4, $\bar{K}$≈8, dt=0.02). Its functional dependence on these parameters is not established here; back-of-envelope arguments for the scaling are given in Section 6 but require a systematic parameter sweep to validate. (vi) The $c \approx 0.30$ result holds within the linear-rotation regime and is the significance-based threshold at two interior speeds. The slow condition exhibits a genuine sign reversal ($L_{\max} \approx L^*$); the mid condition's $L_{\max}$=3 marks statistical detectability rather than coordination collapse ($L^* \in (30, 45)$). The quantitative constant c≈0.30 is primarily anchored on the slow condition, where the two operationalisations agree. (vii) The mechanism isolation (Figure 3) directly measures inter-agent angular dispersion and shared-arm angular bias. It does not measure spatial trajectory overlap, spatial coverage, or provide a formal mediation chain from angular dispersion to capture counts; that causal chain is consistent with the evidence but not directly tested. (viii) The discrepancy between the theoretical $L^* \propto (\omega \cdot dt)^{-2/3}$ and the observed $L_{\max} \propto (\omega \cdot dt)^{-1}$ is attributed to policy nonlinearity, but the functional form of that nonlinearity is not derived or measured. This exponent discrepancy is an open problem. (ix) The four-regime taxonomy is data-derived, not pre-specified. A predictive rule for regime membership from (ω, H) alone — the central open question of this work — has not been established.

7 Conclusion

We provided evidence for a stale-coordination failure mechanism in which the shared arm drives all collectors toward a systematically stale direction, reducing the inter-agent angular diversity that makes coordination beneficial. Instrumented runs directly show the mechanism's two directional signatures (Figure 3); the causal chain from reduced angular diversity to reduced captures is supported by the performance crossover but not isolated by a formal mediation analysis. The coordination benefit landscape under rotation is *regime-dependent*: different rotation speeds produce qualitatively different phenomena, and identifying which regime applies is as important as the within-regime scaling constant.

Four regimes emerge from this study. *Linear-rotation regime* ($\omega L_{\max} dt \approx 0.30$ rad): two interior speeds ($\omega \in \{1.5, 5.0\}$) show a clean significance-based crossover at $L_{\max}/T_{\mathrm{corr}} \approx 0.30$; the bias/noise-parity theory predicts $L^* \propto (\omega \cdot dt)^{-2/3}$ (a different functional form), with the empirical crossover earlier because the relevant threshold is policy nonlinearity, not directional MSE parity. *Sinc-attenuation regime* ($\omega \in \{2.0, 2.5\}$): amplitude attenuation (sinc$\in [0.59, 0.87]$ at large L) weakens the mechanism across the entire tested L range, suppressing the sharp crossover and producing broad benefit instead. *Rapid-rotation regime* (ω=10.0): predicted $L_{\max}$ falls below integer resolution; the only nominally significant cell (L=30, t=2.04) is a false positive at 4.7% estimate amplitude; the mechanism is present but undetectable at n=32. *Long-episode regime* (ω=1.0, T_{corr}≈H): only L=1 significant; mechanism unresolved.

Within the linear-rotation regime and the specific parameter setting studied (α=0.06, σ=0.06, M=4, dt=0.02, sliding-window controller), the heuristic $L \lesssim 0.30/(\omega \cdot dt)$ avoids the stale-coordination failure. The $c \approx 0.30$ constant rests on two interior anchors; the borderline slow anchor (p=0.049) was confirmed at n=64 in a prespecified sensitivity analysis (t=+1.87, p<0.05, df = 63). The central open question is what determines which regime a given (ω, H) pair falls into — equivalently, what governs whether a sharp coordination crossover exists at all.

A Full Per-Cell Results

Table 7 reports $\bar{\Delta}$, SE, and p-values for all window-method cells. Decay-method $L_{\max}$ results are summarised in Table 6 in the main text. Significance at p<0.05 is marked with (*).

B Reproduction Gate Details

Gate run: ω=0, L=1 (stateless sliding window), seeds 6001–6032. $\bar{\Delta}$=+1.188 (SE = 0.213, 95% CI [+0.77, +1.61]); one-sided t-test: t=5.58, df=31, p<0.001; sign count 20/32. This matches SPS-C03's $\bar{\Delta}$=+1.19 (SD 2.44, 20/32 positive) within rounding error, confirming the rotating-field implementation is correct at ω=0. Note: the reproduction gate uses L=1, not L=H=67. The L=67 stationary cell in Table 2 is a separate full-history sliding-window cell that coincidentally yields $\bar{\Delta}$=+1.19 because field stationarity makes window length irrelevant; it is *not* the gate cell.

C Proof Sketch: SNR under Angular Bias

At step t, the window estimate $\hat{v}(t, L)$ (Equation 1) is the mean of L observations at steps $t, t-1, \ldots, t-L+1$. Its expected direction is $\hat{\theta} = \theta(t) - \omega(L-1)\,dt/2$ (lag to the window midpoint) and its sinc-attenuated magnitude is $\alpha\,\mathrm{sinc}(\omega L\,dt/2)$ (the geometric-sum amplitude). Note that the direction lag uses $(L-1)$ (the span from the oldest to the newest observation) while the sinc argument uses L (both $\sin(L\omega dt/2)$ and the denominator L arise from the

Table 7: Window method: all 40 primary non-gate cells plus ω=10.0 exploratory cells (n=32 each).

ω	L	$\bar{\Delta}$	SE	p
very_slow	L1	+0.53	0.44	0.115
	L3	+0.47	0.67	0.243
	L10	+0.22	0.48	0.322
	L30	+0.41	0.40	0.156
	L67	+0.78*	0.46	0.044
slow	L1	+1.47*	0.51	0.002
	L3	+1.50*	0.41	0.000
	L5	+1.66*	0.45	0.000
	L10	+0.94*	0.57	0.049
	L20	−0.38	0.47	0.788
	L30	−0.28	0.52	0.705
	L45	−0.47	0.59	0.786
	L67	−0.44	0.63	0.758
mid	L1	+0.88*	0.39	0.012
	L3	+0.91*	0.34	0.004
	L5	+0.22	0.41	0.296
	L10	+0.25	0.54	0.320
	L20	+0.56	0.46	0.113
	L30	+0.16	0.50	0.378
	L45	−0.25	0.41	0.727
	L67	+0.03	0.43	0.471
fast	L1	+0.47	0.44	0.145
	L3	+0.47	0.44	0.145
	L5	+0.78	0.39	0.023
	L10	−0.09	0.52	0.572
	L20	+0.09	0.46	0.420
	L30	−0.38	0.49	0.777
	L45	+0.13	0.53	0.406
	L67	+0.25	0.50	0.310
ω=10.0	L1	+0.56	0.44	0.107
	L3	+0.22	0.39	0.290
	L5	+0.50	0.47	0.149
	L10	+0.41	0.45	0.185
	L20	+0.44	0.42	0.155
	L30	+1.03*	0.51	0.025
	L45	−0.19	0.63	0.616
	L67	−0.16	0.55	0.612

geometric sum, not the midpoint approximation). The direction formula and the sinc-attenuation approximation are valid only while the sinc factor is positive, i.e. $\omega L\, dt/2 < \pi$ (equivalently $L < \pi/(\omega\, dt)$). In the crossover region (cells at or below $L_{\max}$, where $\omega L\, dt \le 0.30$ rad) the argument is at most 0.15 rad, giving $\mathrm{sinc}(0.15) \approx 0.996$ (< 0.5% amplitude loss), so sinc attenuation is negligible and the direction formula holds. Beyond $L_{\max}$, and in particular at very large L, the sinc factor can become negative: at (ω=5, L=67) the argument is 3.35 rad and $\mathrm{sinc}(3.35) \approx -0.06$. A negative sinc factor means the expected averaged vector *flips direction by* π, so the directional formula no longer holds for such cells. No mechanistic claim is made here about the large-L regime; it is outside the scope of this analysis, and those cells are empirically in the degraded region ($\bar{\Delta} \le 0$). The dominant factor in the crossover region is the angular lag $\beta = \omega(L{-}1)\, dt/2$.

The SNR of the direction estimate for M agents is approximately

$$\mathrm{SNR}_{\mathrm{shared}}(L, \omega) = \frac{\alpha \cos(\beta)}{\sigma/\sqrt{M\bar{K}L\, dt}},$$

and the ratio to the independent arm is $\sqrt{M}$, independent of L and ω — showing the simple model cannot predict negative $\bar{\Delta}$. Full derivation available on request.

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